Nutritional Considerations for the Power Athlete

Jennifer Payne, MD, CAQSM, FAAFP;[1] McKenna Welshans, MBA, RD, CSCS;[1] Shannon Seeland, MD;[2] and Meagan Nielsen, MS, RD, CSSD, LDN[3]

Abstract

Nutrition is a significant factor in a power athlete's ability to achieve hypertrophy, strength, lean body mass, and overall performance goals. Optimizing an athlete's macronutrient balance, timing of intake, and hydration is essential for advancing performance and should be fluid as the athlete transitions between cycles of training, matching nutrient requirements to intensity of training throughout periodization. Supplement use can help athletes meet their performance and nutrition goals when used as an adjunct to a well-chosen diet, both by direct ergogenic effect and by reducing risk of illness or injury. Educating athletes and coaches on an optimal nutrition plan to support training, performance, and health is critical to prevent the negative effects that may come from poor diet, dangerous weight cutting practices, and relative energy deficiency in sport.

Introduction

While many athletes and coaches are aware of the importance of nutrition and hydration on performance, there exists a culture in many power sports of supplementation over whole foods and increased protein intake sometimes to the exclusion of other nutrients. This article seeks to simplify and summarize goal-directed nutritional strategies for the power athlete, as well as provide guidance for those looking to build muscle mass/power.

What Is a Power Athlete?

We begin by defining "power." Scientifically, this is work divided by time. Work is force over distance. Force, as we know, is mass multiplied by acceleration. Thus, a power athlete is one that moves a mass quickly over a distance. Examples of power sports include weightlifting, gymnastics, sprinting, and throwing/jumping events. We will be discussing considerations for fueling these athletes (1).

For power athletes, energy expenditure happens through a combination of pathways, and considerations for fueling will depend on where the athlete is in their training cycle, as well as training goals. Energy pathways for adenosine triphosphate (ATP) generation include phosphagen breakdown, nonoxidative glycolysis (anaerobic metabolism), and oxidative phosphorylation (aerobic metabolism). Power athletes will need a combination of fuel and muscle fiber types to achieve training goals.

Carbohydrate is the main fuel for oxidative phosphorylation, as well as nonoxidative glycolysis; fat is only broken down in oxidative phosphorylation (aerobic metabolism). There often can be a high lactate generation with power sports, thus, the athletes need to have muscle with a strong buffering capacity.

Resting Metabolic Rate

The resting energy expenditure (REE) is the amount of energy the body uses to maintain normal functions and homeostasis. REE is affected by factors, such as age, body composition, sex, climate, hormonal status, and body temperature (2).

$$\text{(RMR) kcal} \cdot \text{d}^{-1} : \text{(males)} = 9.99 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} - 4.92 \times \text{age(years)} + 5;$$

$$\text{(RMR) kcal} \cdot \text{d}^{-1} : \text{(females)} = 9.99 \times \text{weight (kg)} + 6.25 \times \text{height (cm)} - 4.92 \times \text{age (years)} - 161.$$

Of note, these authors are not aware of resting metabolic rate (RMR) calculations available for transitioned or transitioning transgender athletes. This is a developing area in sports and an area for further research.

Macronutrients

The overarching goal of sports nutrition is to consume enough energy and the right balance of macronutrients to optimize recovery from exercise (3). This varies depending on factors,

[1]Penn Medicine Lancaster General Health Physicians Sports Medicine, Lititz, PA; [2]Penn Medicine Lancaster General Health Family Medicine Residency, Lancaster, PA; and [3]Plate to Platform, LLC, Chicago, IL

Address for correspondence: Jennifer Payne, MD, CAQSM, FAAFP, Lancaster General Sports Medicine, 51 Peters Rd Suite #200, Lititz, PA 17543; E-mail: jennifer.payne@pennmedicine.upenn.edu.

1537-890X/2107/224–231
Current Sports Medicine Reports
Copyright © 2022 by the American College of Sports Medicine

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

including exercise intensity, duration, frequency, and individual goals. Power sports emphasize hypertrophy, strength, and maintenance of lean body mass depending on the phase of training. Nutritional intake strategies should be adapted to match the current goal while consistently considering recovery of muscle glycogen stores and protein synthesis (1).

A variety of energy sources are used in power sports, with stored ATP and the phosphocreatine system being the primary sources in shorter, more intense movements. Muscle glycogen utilization increases with duration. This varies significantly throughout the phases of training, and nutritional intake should follow this training pattern accordingly. One of the more common and efficient forms of training for strength athletes is linear periodization, which consists of an early period of high-volume training to focus on hypertrophy and a later period of shorter-duration, high-intensity training to emphasize neural adaptations (4).

During the hypertrophy phase, the focus is a high training volume (~5 to 12+ $h \cdot wk^{-1}$) of lower-intensity training to augment aerobic development. Nutrition is centered on a high caloric intake to support this higher training volume. Carbohydrate, fat, and overall caloric intake should be highest in this phase. Enhancing neural adaptation involves a slightly lower volume (4 to 10 $h \cdot wk^{-1}$) and higher training intensity. Overall calorie needs tend to decrease slightly in this phase, which is reflected in slightly lower carbohydrate and fat recommendations while maintaining protein intake. The taper/competition phase of training incorporates 3 to 8 $h \cdot wk^{-1}$ of high-intensity, sport-specific exercises, and the nutrition goal is to support this while avoiding weight gain. The decreased training volume requires less overall caloric intake, and thus, fat is decreased, since carbohydrate is still important during the high-intensity training and protein is crucial for augmenting muscle recovery. The transition phase allows for both physiological and psychological recovery, as well as prevents overtraining. During this period of such low volume, nutrition recommendations tend to align more closely with those of the general population (although protein is still slightly elevated since many of these athletes do not cease training altogether), and some minor weight gain is expected during this time (1).

Carbohydrate

Studies are inconclusive as to whether a high carbohydrate intake is as beneficial in power sports as it is in endurance sports. A significant intake of carbohydrate leads to augmented glycogen stores. These show a longer time to exhaustion than after a low-carbohydrate diet (1). Increased carbohydrate intake creates a hormonal environment that reduces protein breakdown by increasing circulation of the anabolic hormone insulin and inhibiting the catabolic hormone cortisol (3). Consistent training in a carbohydrate-depleted state has the potential to compromise immunity and increase the risk of burnout. Power athletes tend to have a higher weight and musculature compared with other types of competitors. This results in lower relative energy needs, which should reflect in their macronutrient intake (5). To balance carbohydrate needs with preventing excess weight gain from a higher carbohydrate intake than is necessary, a moderate carbohydrate recommendation of 4 to 7 $g \cdot kg^{-1}$ BW^{-1} d^{-1} could be beneficial for highly muscled athletes. Timing of this intake is important, and studies have indicated that 1 $g \cdot kg^{-1}$ BW prior to exercise and 0.5 $g \cdot kg^{-1}$ BW during a training session can increase total work capacity, while 1 to 1.2 $g \cdot kg^{-1}$ BW immediately postworkout combined with protein intake optimizes restoration of muscle glycogen and protein metabolism, suggesting a reduction in muscle damage (5).

Protein

Because the primary goals of power sports involve muscle hypertrophy and recovery, consistent protein intake is crucial. The current recommendation is roughly 1.5 to 1.7 g protein·kg $BW^{-1} \cdot d^{-1}$ and exceeding the upper range of these guidelines appears to offer no additional benefit, but instead merely increases amino acid catabolism and oxidation of protein (1). Timing of protein intake also is important to optimize muscle protein synthesis. Evidence shows that breakdown of muscle protein may be decreased with a wider distribution of daily protein intake than with a one-time bolus of daily protein. Studies also indicate that in the case of many athletes, the majority of daily protein intake is obtained at main meals, with very little consumed during in-between snacks, especially those around workouts. It is important to incorporate protein sources that are easily digested shortly before and after exercise to encourage muscle protein synthesis. In general, aiming for roughly 20 g of protein per snack or meal that includes 8 to 10 g of essential amino acids spread out over five to six meals per day can help maximize muscle protein synthesis (5).

Fat

Increased fat intake is encouraged in the hypertrophy phase, where a higher volume of endurance-based training necessitates increased caloric intake. It is estimated that a daily fat intake of 2 $g \cdot kg^{-1}$ BW is required to replenish intramuscular triacylglyceride after prolonged endurance training in excess of 2 h in length (1). When total energy intake is being reduced during the competition phase, fat is often the primary nutrient to be decreased. However, some amount of fat intake is critical at all phases of training because of its role in absorption of fat-soluble vitamins, as a substrate for synthesis of hormones, and for its contribution to cellular membrane and myelin sheath integrity. When considering long-term health, fat type is important. Studies show that saturated fat intake tends to be elevated in strength-power athletes, which is possibly due to an increased intake of animal foods in an effort to obtain adequate protein (5). Several studies of blood lipids in throwers and lifters also have indicated high levels of cholesterol and triglycerides, as well as low levels of HDL cholesterol, which increases risk of heart disease (3). Thus, it is important to monitor intake of high-cholesterol and high-saturated fat foods to maximize long-term health of these athletes.

Preworkout and Postworkout Fueling

Timing of macronutrient intake is crucial to augment performance, reduce gastrointestinal (GI) distress, and optimize muscle protein synthesis. Preworkout meals should be high in carbohydrate (1 to 4 $g \cdot kg^{-1}$ BW), lower in protein, fat, and fiber. Preworkout meals should be consumed 1 to 6 h prior to competition (5). Studies have consistently shown that for exercise that is very intense and/or lasting between 1 and 2 h, a carbohydrate intake of 30 to 60 g h^{-1} provides adequate energy, as well as prevents cognitive fatigue, which can reduce technical errors and enhance skill development (1). When GI distress is prevalent, even rinsing the mouth with a sweet carbohydrate

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

beverage for roughly 12 s every 7 to 10 min has proven to increase 1-h time trial performance because of its activation of brain regions that are associated with reward (1).

Consuming roughly 1 to 1.5 g·kg^{-1} BW of carbohydrate soon after exercise will elevate muscle glycogen resynthesis. Ideally, this should be consumed in close proximity postexercise along with roughly 0.3 g protein·kg BW^{-1} to optimize recovery. Another way to approach this would be to aim for 0.8 g carbohydrate·kg BW^{-1} paired with 0.4 g protein·kg BW^{-1}·h^{-1} in the 5-h postrecovery period, which strengthens the effect of each macronutrient when compared with individual consumption (5). These strategies are reflected in the Figure, which depicts nutrition for different cycles of training. From a practical standpoint, it is important to use practice sessions to determine which foods are the most easily digested by each individual athlete so there is confidence in food choices leading up to a competition. When traveling, packing convenient and nonperishable food items from home can help ensure that suitable postcompetition fuel is available. Sports nutrition products, while not ideal for everyday health of the nonathletic population, can be beneficial choices to help ensure adequate nutrition status when on-the-go. Experimenting during training and planning ahead for travel can ensure successful adherence to a sports nutrition strategy so performance and recovery can be optimized (Table).

Hydration and Electrolytes

Maintaining adequate body water and balanced electrolyte levels is crucial to maintaining performance and health. Dehydration increases core temperature, heart rate, and perceived exertion responses during exercise, especially in hot environments (7). The magnitude of the performance decrement varies based on such factors as environmental temperature, exercise task, and an individual's tolerance to dehydration. Many studies show that acute dehydration is modestly harmful to sport performance involving strength and power, and thus this is something to be considered by these athletes (8).

Total body water and overall hydration are normally maintained within a relatively narrow range by the body (1% hyperhydration and 3% hypohydration), and fluid intake practices in response to physical activity have the potential to collaborate with the body's natural ability to regulate itself or compromise it (9). By retaining optimal hydration status within ±1% body weight, one helps to augment athletic performance, maximize the transfer of metabolic heat, maintain mood, facilitate recovery from exercise, and maintain

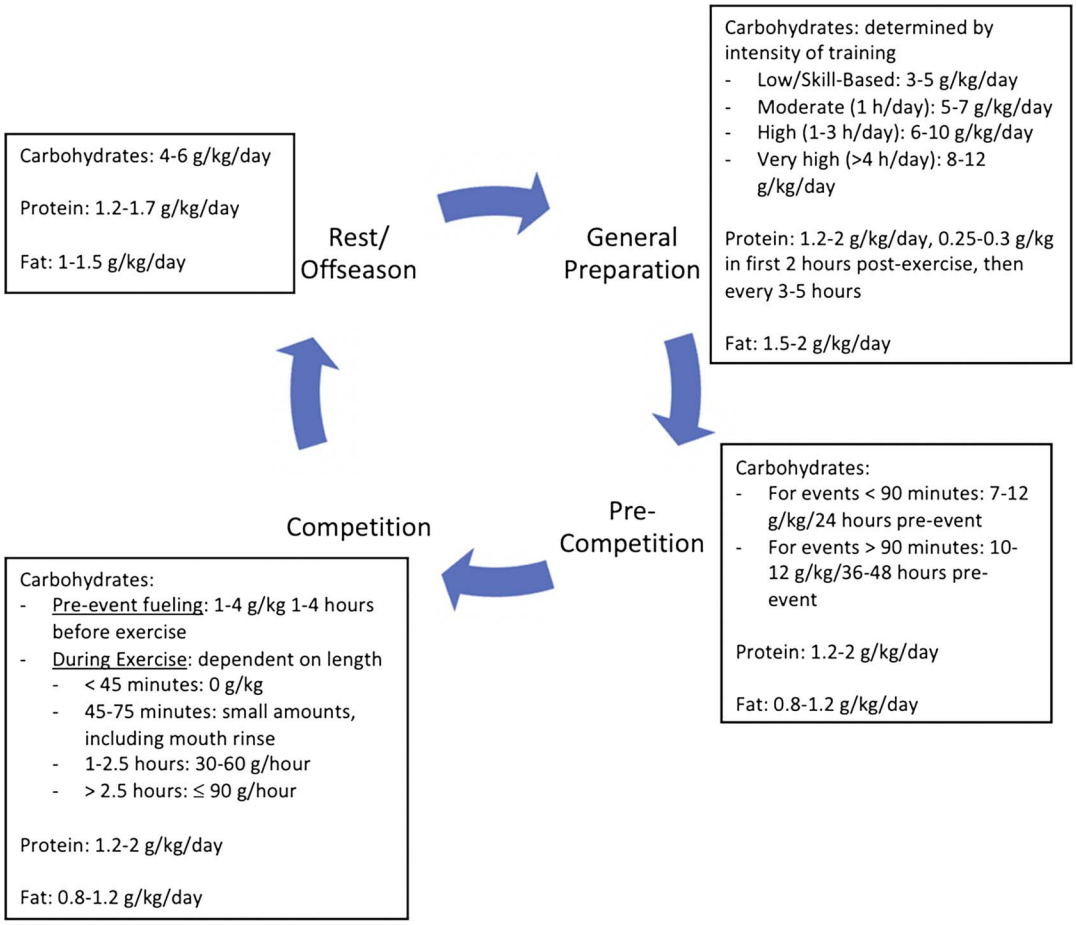

Figure: Macronutrient periodization in cycles of training. Goal breakdown of carbohydrates, protein, and fat are described based on phase of training, including general preparation, precompetition, competition, and rest/offseason. Carbohydrate recommendations specifically are highly dependent on intensity, type, and length of training (1,6).

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Table.
Nutrient needs in different phases of power training (1,4).

Training Phase	Time	Calories	Carbohydrate	Protein	Fat
High-volume hypertrophy	5 to 12 h·wk^{-1}	50 to 70 kcal kg BW^{-1}	6 to 12 g kg BW^{-1}	1.5 to 1.7 g kg BW^{-1}	1.5 to 2 g kg BW^{-1}
High-intensity neural adaptation	4 to 10 h·wk^{-1}	45 to 65 kcal kg BW^{-1}	6 to 10 g kg BW^{-1}	1.5 to 1.7 g kg BW^{-1}	1 to 1.5 g kg BW^{-1}
Taper/competition	3 to 8 h·wk^{-1}	40 to 60 kcal kg BW^{-1}	6–10 g kg BW^{-1}	1.5–1.7 g kg BW^{-1}	0.8–1.2 g kg BW^{-1}
Transition	2 to 4 h·wk^{-1}	25 to 45 kcal kg BW^{-1}	4 to 6 g kg BW^{-1}	1.2 to 1.7 g kg BW^{-1}	1 to 1.5 g kg BW^{-1}

cardiovascular function. Heat loss through evaporation of sweat is the most effective way to thermoregulate, especially in hot conditions. If this is compromised from inadequate availability of fluid, core temperature will increase unnecessarily. Studies show that 50% of people begin their workout hypohydrated, and as little as 2% of body weight loss from hypohydration can affect performance. Thus, it is important to create an individualized strategy for each athlete to avoid this as much as possible (9).

Assessment of Hydration Status

There are several practical ways for athletes to gauge their hydration status. Determining average euhydrated weight can be done by measuring weight in the morning after initial urine void over a period of at least 3 d. Calculating body mass change compared to these euhydrated values over the course of 24 h can help monitor hydration status throughout the day in addition to the personal cues of thirst, void frequency, and urine color (9). When using sweat rate testing to develop a personalized hydration plan, it should be performed at a similar intensity and environment as competition for maximal accuracy. The following is a sweat testing method recommended by the Gatorade Sport Science Institute (10). Prior to a training session, the athlete should void the bladder and measure body mass wearing minimal clothing. *Ad libitum* fluid and food intake can be measured by weighing the bottles/foods before and after consumption. Any urine/feces excreted during the activity should be measured as well. After the exercise session, athletes should towel dry themselves and reweigh wearing the same clothing. Acute body mass change represents 1 mL (0.03 oz) of water (sweat) loss per 1 g (0.002 lb) of body mass loss. The Gatorade Sports Science Institute uses the following equations to calculate whole-body sweat loss and whole-body sweating rate:

$$\text{Equation 1 WBSL(L)} = [\text{Body Mass}_{\text{PRE-EX}} - (\text{body Mass}_{\text{POST-EX}} - \text{Fluid Intake}_{\text{EX}} + \text{Urine Output}_{\text{EX}})].$$

$$\text{Equation 2 WBSR (L/h)} = \text{WBSL/exercise duration}.$$

EX is during exercise, PRE-EX is preexercise, POST-EX is postexercise, WBSL is whole-body sweat loss, and WBSR is whole-body sweating rate (10).

It is common in power sports that involve weight classes to undergo drastic intentional periods of dehydration for the sake of making a lower weight class, only to be followed up with rapid rehydration prior to the event. Studies on the effects of these practices on neuromuscular performance have not been significantly conclusive, but it appears that maximal isometric and eccentric strength could suffer some impairment if adequate hydration status has not been returned prior to the event (11).

Preexercise Hydration

The goal of a hydration strategy should be to begin the exercise session euhydrated and prevent a body mass loss of more than 2% by the end of the exercise period. Care also must be taken not to consume hypotonic liquids in excess that would encourage exercise-associated hyponatremia, which becomes increasingly likely with increased exercise duration/intensity. Beginning a workout hyperhydrated has not been shown to provide physiological or performance benefits and generally is not encouraged (except in instances to prolong the onset of dehydration in endurance events or when fluid availability is limited during the event) due to the risk for developing exercise-induced hyponatremia (9). Because of the variety of factors influencing sweat rate, there must be some flexibility in the hydration routine to accommodate them in any given situation. When preparing for exercise, it is important to space out fluid intake over the course of several hours to enable fluid absorption and prevent hyponatremia and the need to void during the activity. Generally, consumption of ~5 to 7 mL kg^{-1} BW at least 4 h before the task can accomplish this (7). If dark urine or lack of urine is noticed prior to exercise, it may be beneficial for the individual to add another ~3 to 5 mL kg^{-1} BW fluid intake roughly 2 h prior to the event. By consuming beverages with about 20 to 50 mEq·L^{-1} of sodium or small amounts of sodium-containing foods prior to an event, thirst can be stimulated and retention of the consumed fluids can be augmented as well.

Hydration during Exercise

During exercise, onset of thirst signals tends to occur around 2% body mass loss and often tends to be ignored. Thus, relying on thirst signals as a guide for maintaining hydration throughout a workout is not adequate to optimize performance. Assessing average sweat rate enables an accurate midexercise hydration plan to be developed to match the rate of sweat and electrolyte loss rather than relying solely on delayed thirst signal. However, drinking according to thirst can be a safe way to stay hydrated without risking hyponatremia when sweat rate is unknown (9). The longer the duration of a workout, the more important accurate electrolyte and fluid replenishment becomes. However, with exercise bouts lasting an hour or less, water alone should be adequate. Athletes participating in intense exercise lasting longer than an hour may benefit from adding carbohydrates and/or electrolytes to their rehydration fluids, especially in very hot

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

environments. No evidence shows that athletes are helped by sodium supplementation beyond individual losses.

One of the single most important factors to consider is the volume of fluid in the stomach. Maintaining 400 to 600 mL in the stomach, consuming a cooled solution of 3% to 8% concentration, and preventing dehydration of 4% body mass will maximize gastric emptying and prevent it from slowing. The Institute of Medicine for sports beverages more specifically recommends ~20 to 30 mEq L^{-1} of sodium chloride, ~2 to 5 mEq L^{-1} potassium and ~5% to 8% carbohydrate, especially for prolonged physical activity in hot weather (7). By including carbohydrate in the fluid mixture, it delays the onset of low energy availability during longer bouts of exercise. The greatest rates of carbohydrate delivery can be achieved with a mixture of glucose, sucrose, fructose, and maltodextrine (7). Intensities above 80% $\dot{V}O_2$ max also may decrease gastric emptying and intake must be adjusted as necessary to account for this. Generally, ingesting ~200 mL of fluid every 15 to 20 min as able is an effective strategy to maintain hydration without compromising GI function (9).

Postexercise Hydration

The goal of postexercise hydration is to replace any fluid and electrolyte deficit. Ideally, body fluid losses related to exercise should be replaced within a short timeframe. This becomes more important if additional exercise bouts are planned within the same day (7). Aggressiveness of a rehydration routine depends on the magnitude of the deficit, as well as the amount of recovery time available. Body mass losses related to sweat generally require 100% to 150% of calculated fluid losses due to fluid bolus-initiated diuresis, especially during times where recovery time is limited before the next exercise bout (less than 4 h). Another way to measure this is consuming ~1.5 L of fluid for each kilogram of body weight lost. This should be spread out over as many hours as available between exercise bouts and paired with sodium intake to prevent large amounts being lost through urine output. However, when recovery time is more than 12 h, balance can be achieved by ad libitum food and fluid consumption (9).

Optimal hydration recommendations are not yet available for power athletes because of the limited research done on this population. However, using the abovementioned assessment and hydration strategies for pre/mid/postworkout hydration can be an effective and individualized foundation for promoting athletic performance.

Supplements

The use of supplements has grown significantly in athletes over the past years as a multitude of products have become available and widely marketed. For power athletes, supplements can be a practical way to meet nutrition and performance goals when used as an adjunct to a well-chosen, habitual diet. Athletes should consider efficacy, safety of the product, and compliance with antidoping rules when choosing a supplement (12). Supplements are not routinely regulated and therefore can contain excessive amounts of a substance or hidden or banned ingredients that may be harmful to the athlete. The choice to use a supplement should be considered carefully with consideration of third-party confirmatory testing of products and careful attention to any banned substances (12).

The following supplements have significant evidence to suggest a beneficial effect on performance, whether indirectly through the prevention or treatment of nutrient deficiency, or directly through ergogenic effect. While the data supports a general benefit on performance of the following supplements, the effect on the individual is highly specific and related to multiple variables, including genetics, microbiome, and habitual diet. It is recommended that athletes trial supplements in training prior to competition for this reason (12).

Supplement Use to Prevent or Treat Nutrient Deficiency

While not direct ergogenic agents, micronutrients play a significant role in the health and performance of power athletes. Nutrient deficiency can negatively affect sports performance both through a direct inability to train effectively and indirectly through an inability to stay illness and injury free. The most common deficiencies that require supplementation include iron, calcium, vitamin D, and omega-3, though athletes may benefit from other micronutrient supplementation in specific scenarios (12).

Athletes may be at an increased risk of iron deficiency due to low-iron diets, periods of rapid growth, excess loss in sweat, high altitude training, "foot-strike hemolysis," and most commonly in female athletes, menstruation. Low iron availability can lead to impaired hemoglobin synthesis, causing iron deficiency anemia. This impairs the oxygen carrying capacity of the athlete, directly impacting their aerobic capacity and performance. Recommended daily doses of iron are greater than 18 mg and 8 mg for female and male athletes, respectively (12).

Calcium and Vitamin D are integral to bone health of the athlete, which is particularly important to strength athletes in resistance training and weight loading. Vitamin D also plays a role in normalizing muscle strength and improving muscle function. Deficiency is associated with low bone mineral density and increased risk of stress fractures (12). Deficiency is most often associated with avoidance of dairy products, general energy intake restriction, and minimal exposure to sunlight. A 2016 study by Hildebrand et al. found that among male and female college athletes, decreasing vitamin D levels were associated with reduced muscle strength and power scores while controlling for lean body mass. Specifically, they found reduced performance in vertical jump, shuttle run time, triple hop distance, and one-rep-max squat among vitamin D deficient athletes (13).

Daily calcium and vitamin D requirements for athletes are generally equivalent to the general population, 1000 mg and 800 to 1000 IU, respectively, whether through whole foods or supplements or both (12). Skin exposure to sunlight accounts for up to 90% of total vitamin D, therefore people with darker skin and those residing in northern regions in the winter are more likely to be deficient (12). In athletes with relative energy deficiency or menstrual dysfunction, supplementation to achieve daily doses of 1500 $mg \cdot d^{-1}$ of Calcium and 1500 to 2000 IU d^{-1} of Vitamin D is recommended to prevent low bone mineral density (12).

Direct Ergogenic Supplements

While many supplements claim to improve performance, there are only a few that have evidence to support the claim. So far, research has demonstrated the efficacy in performance enhancement for caffeine, creatine, nitrate, beta-alanine, and sodium bicarbonate (12). Again, the effect of each of these supplements has wide interperson variability depending on a

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

multitude of factors, and also depends largely on the appropriate protocol of use. The evidence for each of these supplements, as well as specific considerations for each, are discussed below:

Caffeine

Caffeine has become increasingly popular both alone and as part of energy drinks and preworkout, particularly among strength athletes. The IOC consensus statement on dietary supplements and the high-performance athlete (2018) found on review of the literature, a caffeine dose of 3 to 6 mg kg^{-1} body mass taken 50 to 60 min before exercise resulted in >3% performance gains for mean and peak power output in repeat sprint activities (12). On further review in 2021, the International Society of Sports Nutrition concluded that specific to power athletes, research supports that caffeine at 3 to 6 mg kg^{-1} body mass 60 min prior to exercise may enhance single and intermittent sprint performance, with increases in muscular strength as much as 2% to 7%. However, there remains large variability in caffeine's effect on repeated-sprint performance and muscular endurance in the power athlete, requiring further research (14). Both reviews are in agreement, there is evidence that large doses of caffeine greater than 9 $mg \cdot kg^{-1}$ body mass are not associated with increased performance benefit, but are more likely to increase negative side effects, such as anxiety, tremulousness, tachycardia, nausea, and insomnia (12).

Creatine

Research suggests that creatine may enhance maximum isometric strength of single and repeated rounds of high-intensity exercise less than 150 s duration, with the most significant improvements in tasks less than 30 s (12). When taken long-term, creatine tends to be correlated with lean mass gains and overall improvement of muscular strength and power. Unlike caffeine, athletes will not see an immediate response to creatine for athletic performance. Benefits to performance are found with prolonged and consistent creatine supplementation (12). Suggested dosing is as follows:

Loading phase (research is mixed as to whether this phase is necessary): ~20 $g \cdot d^{-1}$ (divided into 4 equal daily doses), for 5 to 7 d.

Maintenance phase: 3 to 5 g d^{-1} (single dose) for the duration of the supplementation period (12)

While creatine is generally very well tolerated in athletes, a notable side effect is a potential weight gain of 1 to 2 kg after loading, most likely as a result of water retention. This may have significant impacts on power athletes who perform in weight-category sports. Also, creatine may cause GI distress in some athletes. An athlete should begin taking creatine early on in a training cycle to allow the body to adjust to the creatine intake. If a weight class needs to be made, typically a strength and power athlete will cease creatine use about 1 week prior to weigh-in to help reduce water retention (12).

Nitrate

Nitrate supplementation is thought to enhance type II muscle fiber function, enhancing high-intensity, intermittent exercise of 12 to 40 min in studies of team sports. However, there does not appear to be any benefit to exercise lasting <12 min (12). This may limit the direct effect on competition events of power athletes due to the length of events; however, it may have an indirect benefit by enhancing longer training sessions with repeated high-intensity workouts. Acute performance benefits generally occur within 2 to 3 h of a nitrate bolus of 5 to 9 mmol or 310 to 560 mg, and prolonged supplementation also appears to enhance performance particularly among highly trained athletes who are less likely to benefit from acute supplementation alone (12).

Sodium bicarbonate

Sodium bicarbonate acts as a buffering agent, particularly helpful in power athletes who are often limited by acid-base disturbances common from high power output over short periods of time. Single dosing of 0.2 to 0.4 g kg^{-1} body mass, consumed 60 to 150 min prior to exercise, is associated with performance benefits of ~2% in short-term, high-intensity sprints around 60 seconds in duration (12). Efficacy is significantly reduced as task duration exceeds 10 min. Athletes should take into consideration that the most common side effect is significant GI upset (12).

Beta-alanine

Large variability exists in the effect of beta-alanine on athlete performance, however small performance benefits of 0.2% to 3% have been found in both continuous and intermittent exercise tasks of 30 s to 10 min. The recommended dose is ~65 $mg \cdot kg$ $body\ mass^{-1} \cdot d^{-1}$, ingested via a split-dose regimen (e.g., 0.8 to 1.6 g every 3 to 4 h) over an extended supplement time frame of 10 to 12 wk (12).

Making Weight/Relative Energy Deficiency in Sport in Power Sports

Many elite athletes focus on body weight and composition as a means to optimize their performance. For weight-category power athletes, such as Olympic weightlifting, optimizing the power-to-weight ratio by increasing strength and minimizing weight gives the athlete an advantage. However, excessive attempts to minimize fat-free mass or body weight can be harmful to the athlete and cause unintended performance deficits. In the short term, "making weight" can, if not done carefully, lead to negative health consequences and impaired performance. In the long term, chronic imbalance of energy input and expenditure can lead to Relative Energy Deficiency in Sport (RED-S), a syndrome that can have lasting effects on nearly every body system.

Making Weight Via Weight Cuts

"Weight cutting" is a very common practice among weight-category sports. Here the aim is to achieve a theoretical performance advantage by being able to compete in a lower weight division than the athlete's typical day-to-day body mass. This is often achieved through acute weight loss strategies including significantly reducing fluid and food intake, increasing exercise load, or inducing excessive perspiration. Athletes reported a range of practices, reducing their energy intake anywhere from ~35% the week before weigh-in to complete restriction on weigh-in day (15). On average among competitive wrestlers, this leads to ~5% weight loss at the time of weigh-in day (15). While the athlete may gain a weight class advantage with this practice, careful attention to the techniques used is necessary to prevent unintentional health or performance deficits.

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Two major factors impacted by weight cutting practices are fundamental to performance outcomes: hydration status and energy availability. Dehydration is commonly targeted acutely prior to weigh-in as a means to cut weight, but at what point is this detrimental to the athlete? Reale et al. (15) found that the performance effect of total body dehydration of 2.8% was reversible if the athlete was able to aggressively rehydrate for 3 h prior to competition. However, athletes with a total body dehydration of 6% or more were not able to completely reverse the effects even 15 h postweigh-in.

Reducing energy intake in the week to hours prior to weigh-in is another commonly used practice for making weight. Franchini et al. (16) found the rapid weight loss from limiting food intake led to decreased short term memory, vigor, concentration, and self-esteem, and was associated with more confusion, fatigue, rage, depression, and isolation. They found excessive weight loss, greater than 5% body mass, was associated with both aerobic and anaerobic performance deficits. Specific to the power athlete, excessive weight-cutting is thought to reduce muscle buffering capacity, deplete glycogen stores, and cause hydroelectrolytic disturbances (16). However, these effects were generally observed when athletes did not have the opportunity to refuel and rehydrate after weigh-in. They found most athletes were able to recover their anaerobic performance within 3 to 4 h of refueling and rehydrating (16).

To minimize performance and health deficits, these authors recommend implementing the following modifications to weight-cutting practices:

1. Minimize dehydration to less than 3% total body dehydration prior to weigh-in
2. Minimize acute weight loss to less than 5% total body mass
3. Avoid dehydration and carbohydrate restriction if competition is less than 3 h from weigh-in
4. Optimize recovery with aggressive rehydration and refueling with carbohydrates after weigh-in

Female and Male Athlete Triads/Relative Energy Deficiency in Sport

Relative energy deficiency is defined as an overall energy deficiency relative to the balance between dietary energy intake and energy expenditure leading to impaired physiological function. Energy availability then, is calculated as energy input minus the cost of exercise energy expenditure relative to fat-free mass (17). In athletes, this is caused by either a reduction in energy input (food) or an increase in exercise load. When the body is chronically working in a scenario of relative energy deficiency, it resorts to reducing energy expenditure in physiologic systems, causing disruption in hormonal, metabolic, and functional pathways.

This phenomenon was originally described in the female athlete triad of low energy availability, menstrual dysfunction, and low bone mineral density. More recently, relative energy deficiency in sport (RED-S) has been described and associated with impairment of metabolic rate, immunity, protein synthesis, and cardiovascular, endocrine, and psychological health (17). There now exists a definition of male athlete triad: energy deficiency, reproductive suppression, and poor bone health. Education and awareness around this phenomenon is particularly relevant to the power athlete, as data suggests that the prevalence of RED-S/female and male athlete triad among elite athletes is 20% and 13% in adult and female adolescents, and 8% and 3% in adult and male adolescents, respectively (17).

Studies suggest that an optimal energy availability for athletes is ~45 kcal·kg fat-free mass^{-1}·d^{-1}, and that significant physiologic system impairments occur at a level of 30 kcal·kg fat-free mass^{-1}·d^{-1} (17). Below this optimal energy availability, performance also is affected. Athletic performance was positively correlated with energy availability in a study of elite gymnasts, and time trial performance was found to decline in elite swimmers with relative energy deficiency over a 12-wk period in comparison with their teammates with ample energy availability (18). Tornburg et al. (18) found that elite endurance athletes with functional hypothalamic amenorrhea related to energy deficiency had decreased reaction time and neuromuscular performance measured as knee muscular strength and endurance compared with their eumenorrheic counterparts.

To prevent these health and performance deficits, these authors make the following recommendations regarding the female and male athlete triads ("Triads")/RED-S:

1. Perform annual screening for Triads/RED-S in both male and female elite athletes
2. Monitor for disordered eating, weight loss, lack of normal growth and development, menstrual dysfunction, recurrent injury or illness, decreased performance, or mood changes
3. Use screening tools such as LEAF-Q (female only) and EDE-Q (male and female)
4. Educate athletes on Triads/RED-S, healthy nutrition, and risks of dieting
5. Reduce the emphasis on weight, instead emphasizing nutrition and health as a means to performance gains
6. Aim for energy availability of ~45 kcal·kg fat-free mass^{-1}·d^{-1}

Conclusions

Fueling the power athlete appropriately can prove a challenging and complex process requiring focus on goals and the different periods of training: rest, general preparation, precompetition, and competition. Athletes may adjust their protein-carbohydrate-fat mixes and caloric intake accordingly for goals of muscle/mass gain or loss as needed/desired. It is noted that persistently remaining at competition body weight is considered unhealthy as it increases the likelihood of sickness and injury.

Regarding hydration, keeping the power athlete properly hydrated is essential to optimizing performance. Previous practices of cutting weight are not optimal and, while some athletes may choose to turn to this at times, a much preferential way of preparing for sport is to remain within about 3% to 5% of your competition weight at all times. Dehydration of as low as 2% to 5% can lead to significant decrements in performance.

Athletes should be encouraged to work with a qualified registered dietitian nutritionist, preferably with a certification in sports nutrition. This may be provided by the team or found by the athlete. We recommend certification through the

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Academy of Nutrition and Dietetics or similar governing body. While this article provides general applications and guidelines for fueling, the individual athlete, environment, and sweat rate should all be taken into consideration.

Coaches should generally not be involved in weighing their athletes or controlling the athlete's weight; this can lead to body dysmorphia and/or obsessions about weight among athletes, as well as the concerning possibility of coaches restricting or providing opportunities to athletes based on weight goals. The older practices of weighing in front of everyone should be discouraged. This may be a challenging recommendation in sports where making weight is important for qualification. These authors suggest that, in these cases, the athlete and coach keep track of weights and plan for any needed adjustments for several days to weeks in advance. The coach should provide a supportive environment and may assist with nutrition-based strategies to achieve goal weight; however, strategies that may induce bodily or psychological harm should be avoided. It is noted that, for many of these methods (sauna, fluid restriction, spitting), the athlete may become dehydrated and with the short interval between weigh-in and competition, may not have time to adequately hydrate prior to competition, thus resulting in decrements in performance. It also is noted that cyclical variations in weight for females based on their menstrual cycles can be anticipated and charted in advance to better predict nutrition strategies. In general, it may be safer and result in better performance for athletes in weight-based sports to maintain their training weight more in the higher midrange for sport to give some room for error. These authors also support withdrawing an athlete from competition rather than sacrificing his or her health or safety to make weight. Surrounding power athletes with a team that supports their growth and development as a human being over their weight or scores will be beneficial both in creating an environment for success, as well as preserving athlete participation in sport, and minimizing issues prevalent among many athletes, including Triads/RED-S and body image issues.

The authors declare no conflict of interest and do not have any financial disclosures.

References

1. Stellingwerff T, Maughan RJ, Burke LM. Nutrition for power sports: middle-distance running, track cycling, rowing, canoeing/kayaking, and swimming. *J. Sports Sci.* [Internet]. 2011; 29(Suppl. 1):S79–89. [cited 2022 Feb 8]. Available from: https://www.tandfonline.com/doi/full/10.1080/02640414.2011.589469.
2. Islamoglu AH, Kenger EB. Nutrition considerations for athletes with physical disabilities. *Curr. Sports Med. Reports.* [Internet]. 2019; 18:270–4. [cited 2022 Feb 8]. Available from: http://journals.lww.com/00149619-201907000-00007.
3. Burke L, Cox G. *The Complete Guide to Food for Sports Performance: A Guide to Peak Nutrition for Your Sport*. 3rd ed. Updated and expanded. Crows Nest (N.S.W): Allen & Unwin; 2010. 522 p.
4. Baker D, Wilson G, Carlyon R. Periodization: the effect on strength of manipulating volume and intensity. *J. Strength Cond. Res.* [Internet]. 1994; 8:235. [cited 2022 Feb 8]. Available from: http://nsca.allenpress.com/nscaonline/?request=get-abstract&doi=10.1519%2F1533-4287(1994)008%3C0235%3APTEOSO%3E2.3.CO%3B2.
5. Slater G, Phillips SM. Nutrition guidelines for strength sports: Sprinting, weightlifting, throwing events, and bodybuilding. *J. Sports Sci.* [Internet]. 2011; 29(Suppl. 1): S67–77. [cited 2022 Feb 8]. Available from: https://www.tandfonline.com/doi/full/10.1080/02640414.2011.574722.
6. Thomas DT, Erdman KA, Burke LM. Position of the Academy of Nutrition and Dietetics, Dietitians of Canada, and the American College of Sports Medicine: nutrition and athletic performance. *J. Acad. Nutr. Diet.* [Internet]. 2016; 116: 501–28. [cited 2022 Jan 4]. Available from: https://linkinghub.elsevier.com/retrieve/pii/S221226721501802X.
7. American College of Sports Medicine, Sawka MN, Burke LM, Eichner ER, *et al.* American College of Sports Medicine Position Stand. Exercise and fluid replacement. *Med. Sci. Sports Exerc.* [Internet]. 2007; 39:377–90. [cited 2022 Feb 9]. Available from: https://journals.lww.com/00005768-200702000-00022.
8. Coleman E, Rosenbloom C. *Sports Nutrition: A Practice Manual for Professionals*. Chicago (IL): Academy of Nutrition & Dietetics; 2012; 507 p.
9. McDermott BP, Anderson SA, Armstrong LE, *et al.* National Athletic Trainers' Association Position Statement: fluid replacement for the physically active. *J. Athletic Training.* [Internet]. 2017; 52:877–95. [cited 2022 Feb 9]. Available from: https://meridian.allenpress.com/jat/article/52/9/877/191439/National-Athletic-Trainers-Association-Position.
10. Baker LB. Sweat Testing Methodology in the Field: Challenges and Best Practices. *Gatorade Sports Science Institute [Internet].* 2017; Available from: https://www.gssiweb.org/sports-science-exchange/article/sse-161-sweat-testing-methodology-in-the-field-challenges-and-best-practices.
11. Pallarés JG, Martínez-Abellán A, López-Gullón JM, *et al.* Muscle contraction velocity, strength and power output changes following different degrees of hypohydration in competitive Olympic combat sports. *J. Int. Soc. Sports Nutr.* [Internet]. 2016; 13:10. [cited 2022 Feb 9]. Available from: http://www.jissn.com/content/13/1/10.
12. Maughan RJ, Burke LM, Dvorak J, *et al.* IOC consensus statement: dietary supplements and the high-performance athlete. *Int. J. Sport Nutr Exerc. Metabolism.* [Internet]. 2018; 28:104–25. [cited 2022 Jan 8]. Available from: https://journals.humankinetics.com/view/journals/ijsnem/28/2/article-p104.xml.
13. Hildebrand RA, Miller B, Warren A, *et al.* Compromised vitamin D status negatively affects muscular strength and power of collegiate athletes. *Int. J. Sport Nutr. Exerc. Metabolism.* [Internet]. 2016; 26:558–64. [cited 2022 Jan 8]. Available from: https://journals.humankinetics.com/view/journals/ijsnem/26/6/article-p558.xml.
14. Guest NS, VanDusseldorp TA, Nelson MT, *et al.* International Society of Sports Nutrition position stand: caffeine and exercise performance. *J. Int. Soc. Sports Nutr.* [Internet]. 2021; 18:1. [cited 2022 Jan 8]. Available from: https://doi.org/10.1186/s12970-020-00383-4.
15. Reale R, Slater G, Burke LM. Acute-weight-loss strategies for combat sports and applications to Olympic success. *Int. J. Sports Physiol. Performance.* [Internet]. 2017; 12:142–51. [cited 2022 Jan 8]. Available from: https://journals.humankinetics.com/view/journals/ijspp/12/2/article-p142.xml.
16. Franchini E, Brito CJ, Artioli GG. Weight loss in combat sports: physiological, psychological and performance effects. *J. Int. Soc. Sports Nutr.* [Internet]. 2012; 9:52. [cited 2022 Jan 8]. Available from: https://doi.org/10.1186/1550-2783-9-52.
17. Mountjoy M, Sundgot-Borgen J, Burke L, *et al.* The IOC consensus statement: beyond the female athlete triad—relative energy deficiency in sport (RED-S). *Br. J. Sports Med.* [Internet]. 2014; 48:491–7. [cited 2022 Jan 8]. Available from: https://bjsm.bmj.com/lookup/doi/10.1136/bjsports-2014-093502.
18. Mountjoy M, Sundgot-Borgen JK, Burke LM, *et al.* IOC consensus statement on relative energy deficiency in sport (RED-S): 2018 update. *Br. J. Sports Med.* [Internet]. 2018; 52:687–97. [cited 2022 Jan 8]. Available from: https://bjsm.bmj.com/lookup/doi/10.1136/bjsports-2018-099193.

Copyright © 2022 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.